

Management of Pulmonary Cement Embolism



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Objectives



- Introduce patient case
- Discuss Pulmonary Cement Embolism (PCE)
 - Introduction
 - Epidemiology
 - Pathogenesis
 - Clinical Presentation and Diagnosis
 - Treatment
- Review patient case
- Highlight key points of PCE

Patient Case



- SR is a 81 y/o female presents to the clinic on 09/08 for follow-up on anticoagulation therapy for the diagnosis of Pulmonary Cement Embolism.
- PMH:
 - Asthma
 - Seasonal allergies
 - Hypertension
 - Osteoporosis
 - Pneumonia
 - COPD
- PSH:
 - Tonsillectomy - Unknown date
 - Hysterectomy - 1972
 - Vertebroplasty - 08/24/2016
- SH: Former smoker: ½ pack daily for 48 years, quit back in 2015. Denies any alcohol or illicit drug use
- Allergies: NKDA

Home Medications

Medication Name	Directions	Start Date
Aclindium bromide (Tudorza Pressair) 300 mcg/actuation inhaler	Inhale 1 puff into the lungs twice daily.	05/17/16
Azithromycin 250 mg	Take 2 tablets by mouth on day 1, then 1 tablet by mouth daily on days 2-5.	09/04/2016
Combivent respimat 20-100 mcg/actuation mist	Inhale one puff 4 times daily.	02/06/16
Advair 500-50 mcg/dose	Inhale one puff by mouth into the lungs twice daily.	05/17/16
Metoprolol tartrate 50 mg	Take 1 tablet by mouth twice daily	02/06/16
Enoxaparin 80 mg/0.8 mL	Inject 0.73 mLs subcutaneously twice daily	08/30/16
Warfarin 3 mg	Take 1 tablet by mouth nightly	09/01/16
Prednisone 20 mg	Prednisone taper over a period of 14 days.	09/04/16

Snapshot from Hospital Admission

A decorative graphic consisting of a white circle with a teal outline, centered below the title and above the list.

- Patient was initially hospitalized on 08/30/16 for acute respiratory symptoms
- Presented with shortness of breath, dyspnea, wheezing, coughing, and sputum production.

Snapshot from Hospital Admission



- Physical Exam:
 - General: (+) malaise/fatigue, (-) chills and fever
 - Respiratory/chest: (+) cough, (+) sputum, (+) SOB, and (+) wheezing. No respiratory distress, no rales, and no tenderness.
 - Neurological: Alert and oriented to person, place, and time.
- Vitals (08/30/16):
 - Tmax: 98.2 F (36.8 C)
 - HR: 108
 - RR: 27
 - BP: 145/71 mmHg
 - Ht: 1.65 m (5'5)
 - Wt: 72.6kg (160 lbs.)
 - SpO2: 95%

Snapshot from Hospital admission



- CBC:

WBC	12.8 (H)
Hemoglobin	13.6
Hematocrit	40.7
Platelet Count	271

- Radiology:

- Chest X-ray: negative
- CT scan: No clot PE, but has segmental/sub-segmental embolism that appears to be vertebroplasty materials. Pulmonary infiltrates throughout lobes on both lungs

- Other:

- D-dimer: **1.57 (H)**

Initial Assessment on 08/30



- **Diffuse Bilateral Pneumonia**
 - No risk factors for HCAP
 - Symptoms and CBC indicative of bacteria pneumonia.
 - Treat for routine pathogens
- **Pulmonary Cement Embolism**
 - Discussed case with pulmonology and pharmacy. Unsure if PCE is causing respiratory symptoms.
 - Anticoagulation is recommended over surgical removal of vertebroplasty material
- **COPD Exacerbation**
 - Scheduled nebulizer, PRN albuterol nebulizer. and O2 supplementation as needed.

Inpatient Therapy 08/30-09/04



- **Pneumonia:**
 - Ceftriaxone 2 g IV Q 24 H
 - Azithromycin 500 mg IV Q HS
- **Pulmonary Cement Embolism:**
 - Enoxaparin 70 mg (1 mg/kg) SQ BID
 - Warfarin 3 mg PO HS (started on 09/01)
- **COPD Exacerbation**
 - Budesonide Nebulization 0.5 mg BID
 - Ipratropium-albuterol (duoneb) 2.5mg/3mL nebulizer solution Q 4H PRN
 - Methylprednisolone 40 mg IV Q 8H

Discharge Medications on 09/04



- **Pneumonia:**
 - Azithromycin 250 mg: 2 tablet PO on day 1, then 1 tablet PO on days 2-5.
- **Pulmonary Cement Embolism:**
 - Patient to be on anticoagulant therapy for 6 months with an INR goal of 2-3.
 - **Enoxaparin 80 mg/0.8mL: Inject 0.73 mL SQ BID**
 - **Warfarin 3 mg PO Q HS**
- **COPD:**
 - Continue current maintenance and PRN inhalers/nebulizers.
 - Prednisone 20 mg taper for 10-14 days then STOP.

Pulmonary Cement Embolism Overview



- Introduction
- Epidemiology
- Pathogenesis
- Presentation and Diagnosis
- Treatment

Introduction



- Pulmonary Cement Embolism (PCE) is a complication of vertebroplasty or kyphoplasty, where injected bone cement leaks and embolizes to the lungs.
- Most cases are asymptomatic.
- Symptoms usually occur within 24-48 hours of procedure.

Vertebroplasty & Kyphoplasty



- Both are minimally invasive procedures to treat vertebral compression fractures (VCF) of the spine.
- Vertebroplasty – Polymethyl methacrylate (PMMA) is injected through a needle into a fractured bone in the vertebral column.
- Kyphoplasty - a balloon is inserted and inflated to expand the vertebra before filling the space with bone cement.

Epidemiology



- There is an estimated 70,000-100,000 vertebroplasty procedures performed each year in the US.
- PCEs occur in about 26% of patients undergoing Vertebroplasty.
- While an estimated 4.6% of kyphoplasty patients experience PCEs.

1. Am Fam Physician. 2002 Aug 15;66(4):611-5.
2. Lung India. 2015 Nov-Dec;32(6):602-5
3. Eur Spine J (2009) 18:1257-1265

Pathogenesis



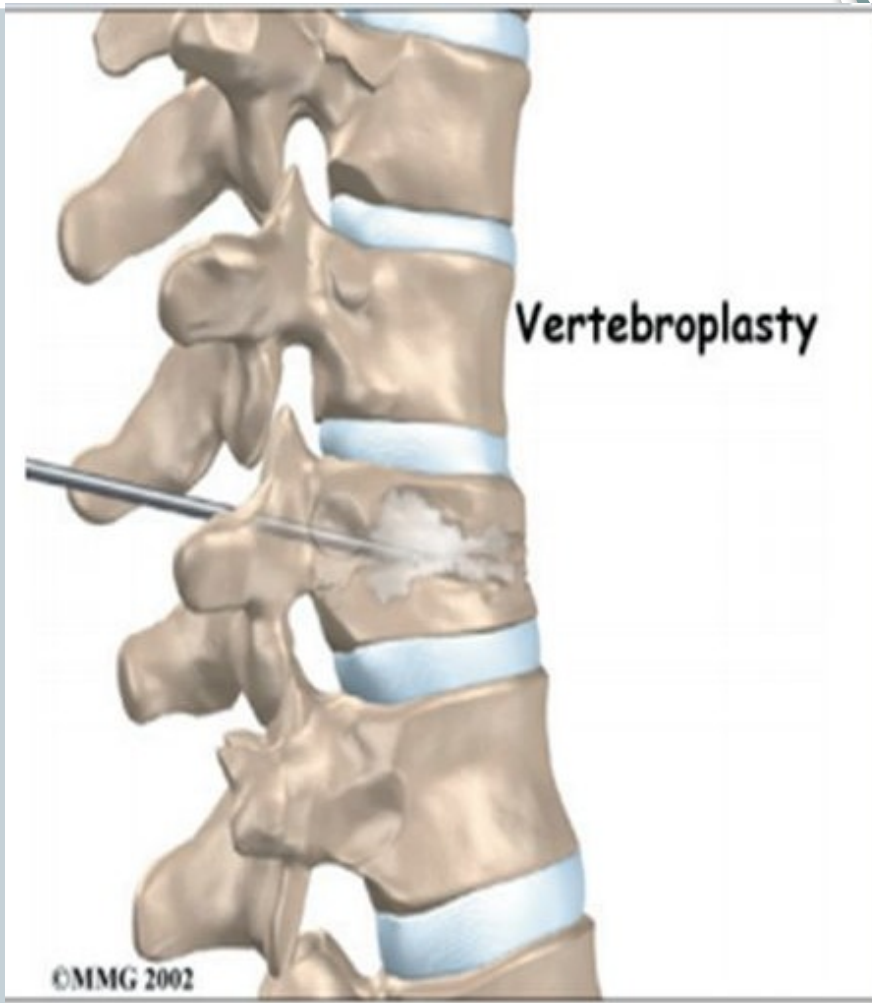
- PCEs are a result of accidental extravasation of bone cement into the the vertebral venous plexus where the cement enters the venous system then migrates to the lungs.
- Extravasation usually occurs due to early injection, injecting too quickly or with too much pressure.

PCE Risk factors



- Cement leakage
- Age >70
- Higher injected cement volumes (> 9 mL)
- Numbers of treated vertebrae per session (>3 vertebrae)
- Type of guidance equipment

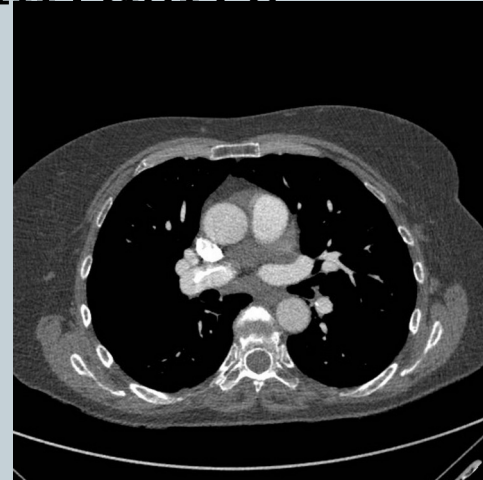
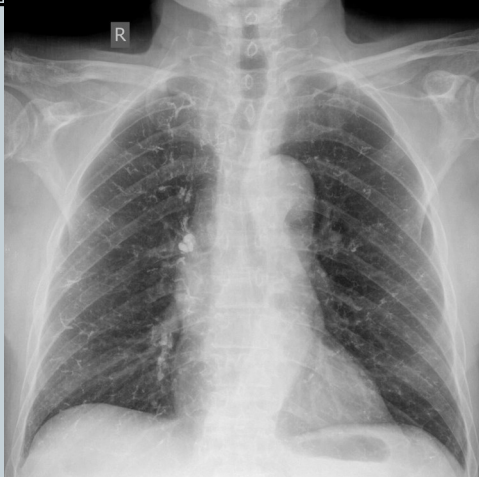
Vertebroplasty vs Kyphoplasty



Diagnosis



- Chest X-ray – shows dense opacities in the lung
- CT scan of the chest – recommended
- Pulmonary angiogram - invasive



1. Orthop Surg. 2012 Aug;4(3):182-9
2. <http://radiopaedia.org/articles/pulmonary-cement-embolism>

Clinical Presentation



- Most cases are asymptomatic, with only 1% of patients being symptomatic.
- Symptomatic patients have presented with:
 - Dyspnea
 - Tachypnea
 - Tachycardia
 - Pleuritic chest pain
 - Hemoptysis

Treatment



- Depends on clinical presentation
- Asymptomatic patients generally only require monitoring and follow-up
- Treatment reduces risk of thrombus formation on cement material
- Treatment options:
 - Surgical Removal
 - Anticoagulation

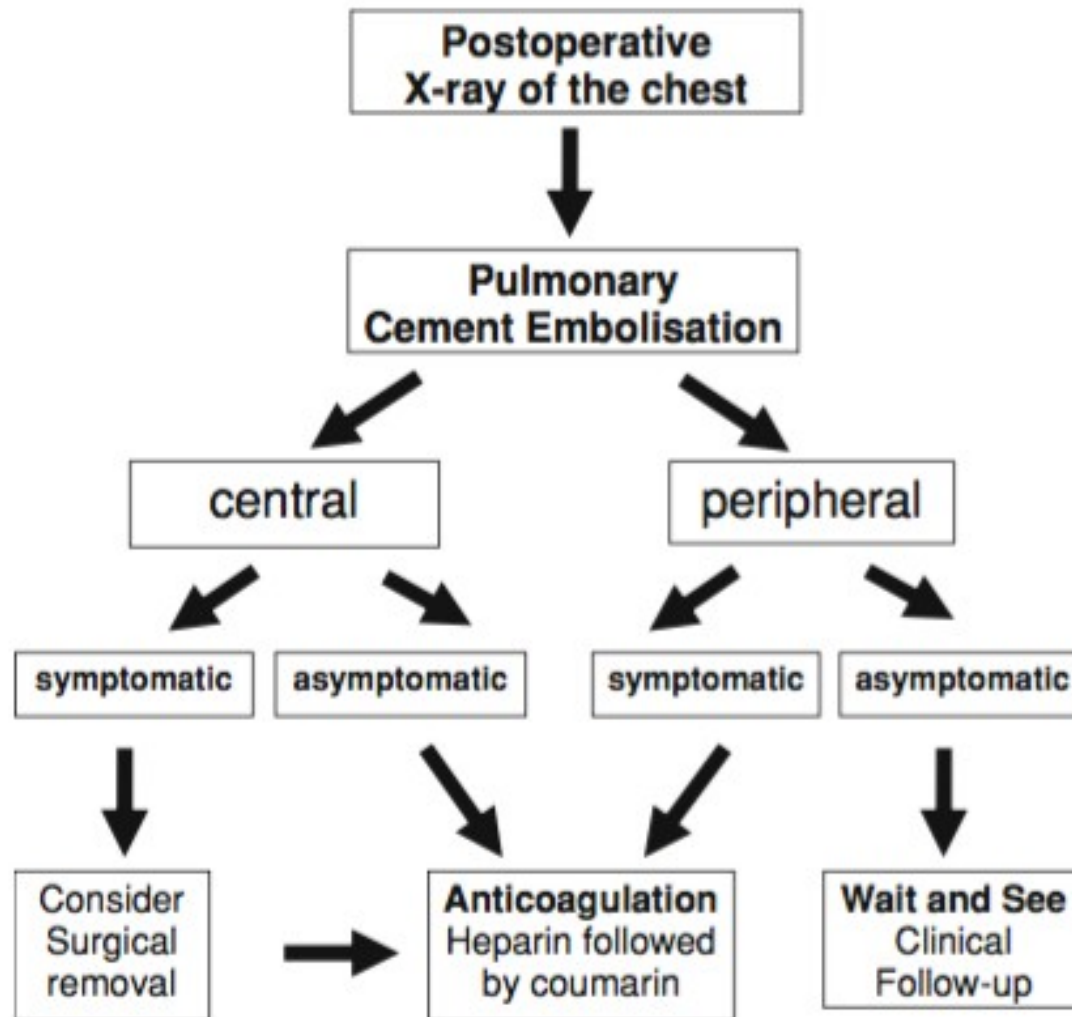


Fig. 1 Decision tree for the management of pulmonary cement embolism as applied in our institution

Surgical Removal vs. Anticoagulation

Surgical Removal

- Reserved for symptomatic central PCE
- Percutaneous preferred over open surgery

Anticoagulation

- Preferred intervention
- Long term follow-up studies are scarce
- Follow standard treatment guidelines for provoked PE
- Duration of therapy is 3-6 months.

Previous Studies



Study	Objective	Methods	Results	Citation
Frequency and Outcome of Pulmonary Polymethylmethacrylate Embolism during Percutaneous Vertebroplasty (2008)	Determine the true incidence of PCE in patients receiving vertebroplasty and determine long-term effects.	Retrospectively assessed the frequency and outcome of PMMA embolism during Vertebroplasty in 299 patients.	- Pulmonary PMMA embolism occurred in 2.1% of patients. - None of the patients developed clinical symptoms during or after procedure. - After 1 year, no symptoms, deaths, or pulmonary reactions were reported among patients.	AJNR Am J Neuroradiol. 2008 Nov;29(10):1983-5
Management of pulmonary cement embolism after percutaneous vertebroplasty and kyphoplasty: a systematic review of the literature (2009)	Evaluate incidence and treatment strategies for the management of PCE.	Literature analysis of 3774 case reports of PCE post-vertebroplasty and kyphoplasty.	- Symptomatic PCEs only occurred in 1.3% of patients. - Only 3.2% of asymptomatic patients received anticoagulation. - 42% of symptomatic patients received anticoagulation, while only 19% of patients received surgical removal.	Eur Spine J. 2009 Sep;18(9):1257-65.
Pulmonary Cement Embolism Associated with Percutaneous Vertebroplasty or Kyphoplasty: A Systematic Review (2012)	Summarize the incidence, clinical features, prophylaxis, and therapeutic management of PCE after vertebral cement reinforcement.	Literature analysis of 5 observational comprising a total of 563 patients.	- Rates of PCE ranged from 4.6-26% among studies. - Most common symptoms were shortness of breath, chest pain, and hypotension. - 2% of patients died due to complications related to procedure (not limited to PCE).	Orthop Surg. 2012 Aug;4(3):182-9

Patient Case Continued



CHEST GUIDELINES



- In patients with a proximal DVT of the leg or PE provoked by surgery, we recommend **treatment with anticoagulation for 3 months** over (i) treatment of a shorter period(Grade 1B), (ii) treatment of a longer, time-limited period (eg, 6, 12, or 24 months)(Grade 1B), or (iii) extended therapy (no scheduled stop date)(Grade 1B).
- In patients with PE who are treated with VKA, **we recommend a therapeutic INR range of 2.0 to 3.0** (target INR of 2.5) over a lower (INR < 2) or higher (INR 3.0-5.0) range for all treatment durations (Grade 1B).
- In acute DVT or PE, we recommend **initial treatment with LMWH, UFH or fondaparinux for at least 5 days** rather than a shorter period (Grade 1C); and initiation of vitamin K antagonists (VKAs) together with LMWH, UFH, or fondaparinux on the first treatment day, and discontinuation of these heparin preparations when the **international normalized ratio (INR) is ≥ 2.0 for at least 24 h** (Grade 1A).

1. *Chest*. 2012;141(2_suppl):7S-47S.

doi:10.1378/chest.1412S3

2. *Chest*. 2012;141(2_suppl):48S-52S.

doi:10.1378/chest.11-2286

Patient Case Cont.



Date:	INR Value
09/01 (baseline)	1.1
09/02	1.1
09/03	1.9
09/04	2.3
09/08 (follow-up visit)	2.6

Patient Case Cont.



- During 09/08 follow-up visit, patient's INR is therapeutic at 2.6. Provider decided to discontinue bridging with enoxaparin and continue warfarin 3 mg PO HS.
- Patient will follow up on 09/21 for routine INR check.

Patient Case Cont.



- Pulmonary Cement Embolism

- Assessment:

- ▮ Since patient initially presented as symptomatic, it is reasonable to start anticoagulation therapy with an INR range of 2.0-3.0.
- ▮ Patient had 2 therapeutic INRs on 09/04 and 09/08 so discontinuation bridging is appropriate.
- ▮ However, the treatment duration for patient's warfarin therapy is 6 months, which is longer than the duration recommended by CHEST guidelines for provoked pulmonary embolism.

Key Points



- Pulmonary Cement Embolism (PCE) usually presents asymptotically and there is limited evidence showing negative long-term outcomes related to PCE.
- Treatment is controversial.
- Anticoagulants and surgical interventions are reserved for symptomatic PCE.
- Symptomatic PCE is generally treated with standard PE therapy.

Questions?

